

Homework 5

EE 999 — Introduction to Control Systems

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Due Wednesday, October 14, 2026

Problem 1.A — Transfer function

$$U(s) = k_p R(s) \quad (1)$$

$$Y(s) = \frac{b_1}{s + a_1} U(s) \quad (2)$$

$$Y(s) = \frac{k_p b_1}{s + a_1} R(s) \quad (3)$$

$$G_{ry1}(s) = \frac{Y(s)}{R(s)} = \frac{k_p b_1}{s + a_1} \quad (4)$$

Problem 1.B — Transfer function

$$Y(s) = \frac{b_1}{s + a_1} U(s) \quad (5)$$

$$U(s) = \frac{k_p s + k_i}{s} R(s) \quad (6)$$

$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \quad (7)$$

$$G_{ry2}(s) = \frac{Y(s)}{R(s)} = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} \quad (8)$$

Problem 1.C.1 — Transfer function

$$Y(s) = \frac{b_1}{s + a_1} U(s) \quad (9)$$

$$U(s) = \frac{k_p s + k_i}{s} E(s) \quad (10)$$

$$E(s) = R(s) - Y(s) \quad (11)$$

$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} E(s) \quad (12)$$

$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} (R(s) - Y(s)) \quad (13)$$

$$\left(1 + \frac{b_1(k_p s + k_i)}{s^2 + a_1 s}\right) Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \quad (14)$$

$$\frac{s^2 + (a_1 + b_1 k_p)s + b_1 k_i}{s^2 + a_1 s} Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \quad (15)$$

$$Y(s) = \frac{b_1 k_p s + b_1 k_i}{s^2 + (a_1 + b_1 k_p)s + b_1 k_i} R(s) \quad (16)$$

$$T_{ry3}(s) = \frac{Y(s)}{R(s)} = \frac{b_1 k_p s + b_1 k_i}{s^2 + (a_1 + b_1 k_p)s + b_1 k_i} \quad (17)$$

Problem 1.C.2 — Assign Values

$$b_1 = 3, a_1 = 2, k_p = 1, k_i = 5$$

$$T_{ry3}(s) = \frac{(3)(1)s + (3)(5)}{s^2 + ((2) + (3)(1))s + (3)(5)} \quad (18)$$

$$T_{ry3}(s) = \frac{3s + 15}{s^2 + 5s + 15} \quad (19)$$

Problem 1.C.4 — Step response

Code: hw_05_1_C_4.m

```

1  if exist('OCTAVE_VERSION', 'builtin')
2      pkg load control;
3  end
4
5  b_1 = 3;
6  a_1 = 2;
7  k_p = 1;
8  k_i = 5;
9
10 num = [b_1*k_p, b_1*k_i];
11 den = [1, (a_1 + b_1*k_p), b_1 * k_i];
12
13 T_ry = tf(num, den);
14
15 T_ry_poles = pole(T_ry)
16 T_ry_zeros = zero(T_ry)
17
18 figure;
19 step(T_ry);
20 grid on;
21 title("Step response of T_{Ry3}(s)");
22
23 [yy, tt] = step(T_ry);
24
25 T_ry_overshoot = max(yy) - 1

```

Output: output/hw_05_1_C_4.txt

```

T_ry_poles =

    -2.5000 + 2.9580i
    -2.5000 - 2.9580i

T_ry_zeros = -5
T_ry_overshoot = 0.1134

```

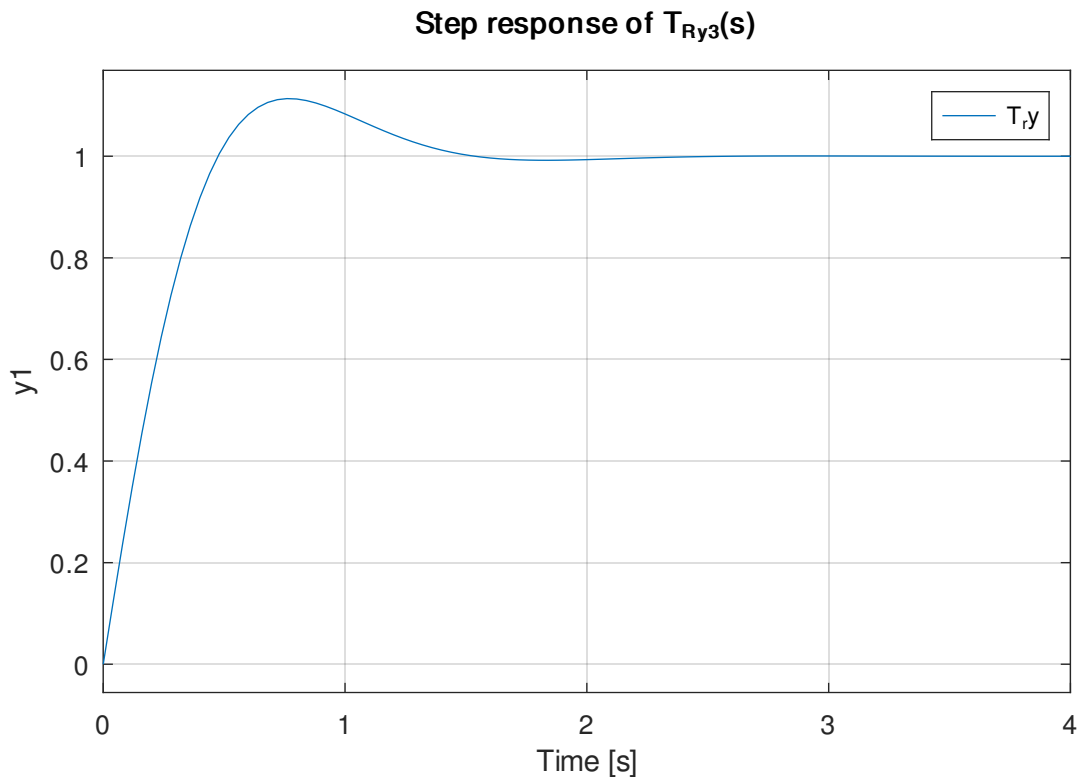


Figure 1: Step response of $T_{ry3}(s)$ with $k_p = 1$, $k_i = 5$.

Problem 1.C.5 — Overshoot

from above code, overshoot is %11.34

Problem 1.C.6 — Adjusted k_p **Code:** hw_05_1_C_6.m

```
1 if exist('OCTAVE_VERSION', 'builtin')
2     pkg load control;
3 end
4
5 b_1 = 3;
6 a_1 = 2;
7 k_p = 1.6;
8 k_i = 5;
9
10 num = [b_1*k_p, b_1*k_i];
11 den = [1, (a_1 + b_1*k_p), b_1 * k_i];
12
13 T_ry = tf(num, den);
14
15 T_ry_poles = pole(T_ry)
16 T_ry_zeros = zero(T_ry)
17
18 figure;
19 step(T_ry);
20 grid on;
21 title("Step response of T_{Ry3}(s)");
22
23 [yy, tt] = step(T_ry);
24
25 T_ry_overshoot = max(yy) - 1
```

Output: output/hw_05_1_C_6.txt

```
T_ry_poles =

    -3.4000 + 1.8547i
    -3.4000 - 1.8547i

T_ry_zeros = -3.1250
T_ry_overshoot = 0.044085
```

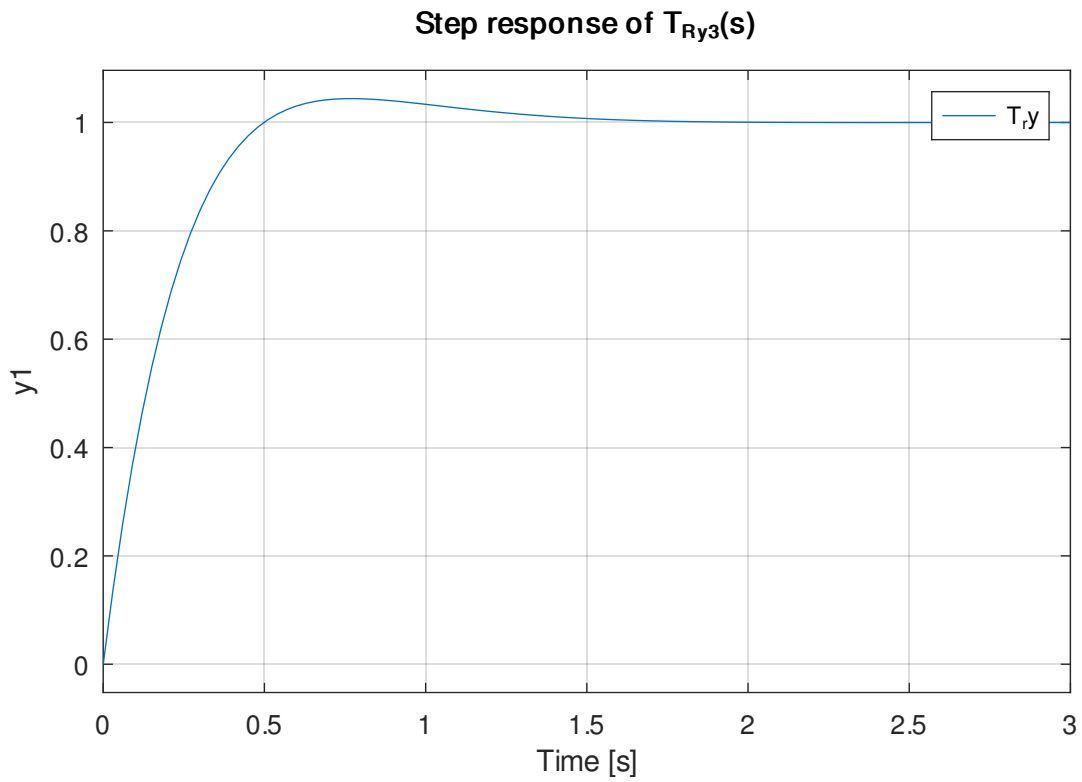


Figure 2: Step response of $T_{ry3}(s)$ with $k_p = 1.6$, $k_i = 5$.

Yes the overshoot is less than 5% at 4.41%.

Problem 2 — Pole and zero locations

Code: hw_05_2.m

```
1 if exist('OCTAVE_VERSION', 'builtin')
2     pkg load control;
3 end
4
5 num_1 = [1 2];
6 den_1 = [1 10 8];
7 G_1 = tf(num_1, den_1);
8 G_1_poles = pole(G_1)
9 G_1_zeros = zero(G_1)
10
11 num_2 = [5 15];
12 den_2 = [1 2 10];
13 G_2 = tf(num_2, den_2);
14 G_2_poles = pole(G_2)
15 G_2_zeros = zero(G_2)
16
17 num_3 = [1 3];
18 den_3 = [1 -4 8];
19 G_3 = tf(num_3, den_3);
20 G_3_poles = pole(G_3)
21 G_3_zeros = zero(G_3)
22
23 num_4 = [5 15];
24 den_4 = [1 2 6 10];
25 G_4 = tf(num_4, den_4);
26 G_4_poles = pole(G_4)
27 G_4_zeros = zero(G_4)
28
29 num_5 = [2 9];
30 den_5 = [1 2 6 30];
31 G_5 = tf(num_5, den_5);
32 G_5_poles = pole(G_5)
33 G_5_zeros = zero(G_5)
```

Output: output/hw_05_2.txt

```
G_1_poles =

    -9.1231
    -0.8769

G_1_zeros = -2
G_2_poles =

    -1 + 3i
    -1 - 3i

G_2_zeros = -3
G_3_poles =

     2 + 2i
     2 - 2i

G_3_zeros = -3
```

```
G_4_poles =  
  
-0.1090 + 2.3664i  
-0.1090 - 2.3664i  
-1.7820 +      0i  
  
G_4_zeros = -3  
G_5_poles =  
  
-3.1365 +      0i  
 0.5683 + 3.0400i  
 0.5683 - 3.0400i  
  
G_5_zeros = -4.5000
```